Parsa Tasbihgou

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Education

- **Ecole Polytechnique Federale de Lausanne (EPFL)** Lausanne, Switzerland
  MSc - Computer Science; GPA: 5.3/6 *September 2023 - Now*
- **University of Tehran** Tehran, Iran
  BSc - Computer Science; GPA: 18.3/20 *September 2019 - August 2023*

Research Interests

- **Cryptography, Isogeny-based cryptography, Lattice-based cryptography**
- **Probabilistic proofs, Zero-knowledge proofs**
- **Algebra, Number theory**
- **Graph theory, Combinatorics**

Research Experience

- **Internship at COSIC, KU Leuven**
  Supervisor: Dr. Wouter Castryck *July-September 2024*
  I visited the isogeny group at the COSIC lab in KU Leuven and I was hosted by Wouter Castryck. During my stay there I studied a variety of problems including: Finding isogenies between ordinary elliptic curves with large prime conductor gap, more generally walking up in a volcano (ordinary or oriented supersingular). Using more efficient orientations to construct class group actions similar to ScallopHD. Quantum secure VDFs. Quantum complexity of finding large prime degree isogenies between oriented elliptic curves. Structure of the isogeny graph of non-simple ordinary abelian surfaces.

- **Master's thesis at COSIC/LASEC**
  Supervisor: Dr. Wouter Castryck *October-March 2025-26*
  I am currently working on my master's thesis under supervision of Dr.Castryck at COSIC, KU Leuven. My general subject of study is isogeny-based cryptography. Within isogeny-based crypto I am working on a couple of ideas but the main focus is understanding and developing the Isogeny Module Action.

Selected Projects

- **Generalized special soundness for Kilian's Protocol**
  Supervisor: Dr. Alessandro Chiesa *December 2024*
  In this project I am studied Kilian's protocol and its relation to different generalizations of special soundness. I showed that for interactive arguments generated by Kilian's protocol Γ-special soundness and k-special soundness are equivalent. Furthermore there are infinite families of PCP for which the corresponding argument is not k-special sound (therefore also not Γ-special sound).
- **Isogeny-based time-release cryptography**
  Supervisor: Dr. Boris Fouotsa *June 2024*
  In this project I studied time-release cryptographic primitives and instantiations based on isogeny of supersingular elliptic curves. I looked at the verifiable delay function proposed by De Feo et al. based on chains of low degree isogenies and pairings and delay encryption introduced by De Feo and Burdges based on the same VDF. I also looked at two quantum-secure VDFs, one based on high degree rational isogenies and Kani's criterion, the other based on the CGL hash function and SNARGs.
  To broaden my view on this field I also looked at some constructions based on squaring in groups of unknown order.

Publications

- **On the active security of the PEARL-SCALLOP group action**: Joint work with Tako Boris Fouotsa, Marc Houben, Gioella Lorenzon and Ryan Rueger. To appear at PQC 2026.

Awards and Honors

- **Third prize in the International Mathematics Competition (IMC) - 2022**
- **Bronze medal in the National Mathematics Competition for university students - 2022**
- **Top 10 teams in University Of Tehran's ICPC - 2019 & 2022**

Presentations

- **IOPP for algebraic geometry codes**: This presentation was a part of the project for the course "Foundations of Probabilistic proofs" by Dr. Chiesa. We studied and presented a paper titled "IOPP for algebraic geometry codes". This paper extends the ideas from the FRI protocol for the Reed-Solomon codes to create a proof of proximity to AG codes. In addition to presenting some parts of the paper, we also gave a brief background on the algebraic geometry knowledge required to understand the paper.
- **A tour in modern cryptography**: This presentation was a two-part lecture that aimed to introduce students to the main goals and definitions of cryptographic primitives, also introduce the main ideas in classical and quantum-secure cryptography.
- **Introduction to Spectral Graph Theory**: A quick introduction to spectral graph theory and important theorems building up to Kirchhof's Matrix-Tree theorem. In this presentation we used spectral graph theory to talk about chromatic number, graph structure and graph polynomials.
- **Spectral Clustering**: I used my previous experience in spectral graph theory to present a brief introduction to a data mining class. This project made me familiar with graph embedding and topology in data scientific applications.

Teaching Experience

- **University Of Tehran**
  Student Assistant
  - **Data Structures and Algorithms**
  - **Combinatorics**
  - **Graph Theory**
  - **Computation Theory**
  - **Artificial Intelligence**
- **EPFL**
  Student Assistant
  - **Algorithms I** Prof. Svenson
  - **Algorithms II** Prof. Svenson

Work Experience

- **NP Labs**
  Cryptography engineer intern *March-September 2025* I

worked at NP Labs for 6 month as a cryptography engineer intern. During this time I participated in designing protocols for compliant privacy preserving blockchains. As the goal is to deploy these protocols for public use, I learned it is important to take grate care of efficiency and compatibility with other parts of the system that is already deployed as any fundamental change to the underlying systems requires a lot of time and is very costly.
Additionally, I implemented parts of these protocol in Rust and tested them. I learned that implementation cryptography algorithms is extremely delicate as one needs to keep in mind both optimization of the code, faithfulness of the code to the algorithm and most importantly robustness against leaks and attacks.
The most interesting and new part of my experience at NP Labs was the logistical side of developing cryptography. I learned that creating cool protocols is only half of the work, securing funding and adoption for the protocols is almost the same amount of work.

- **Mahsan Co**
  Software engineer; C++ and Python *June 2021 - September 2022*
  - **Security engineer**: I worked as a software engineer in the network security team at Mahsan. During this time I focused on design and implementation of a passive and an active monitoring (deep packet inspection) algorithms for encrypted network traffic.
  - **Code review**: I assisted in code review process in my team.